2 information theory & entropy

Problem 2.1 : shannon entropy

a discrete memoryless source produces symbols from the alphabet $\{A, B, C, D\}$ with probabilities $P_A = \frac{1}{2}$, $P_B = \frac{1}{4}$, $P_C = \frac{1}{8}$, $P_D = \frac{1}{8}$

a) compute entropy $H(x)$ → how much knowledge

$$H(x) = -\sum p_i \log_2 P_i \longrightarrow \left.\begin{array}{l} P_A : \frac{1}{2} \log_2 \frac{1}{2} \\ P_B : \frac{1}{4} \log_2 \frac{1}{4} \\ P_C : \frac{1}{8} \log_2 \frac{1}{8} \\ P_D : \frac{1}{8} \log_2 \frac{1}{8} \end{array}\right\} \text{negative sum} = 1.75 \text{ bits/symbol}$$

b) what would the entropy be if all 4 symbols were equally likely?

what do mean

$$H(x) = \sum_{i=1}^{4} \frac{1}{4} \log_2 \frac{1}{4} = 0.5 \times 4 = 2 \text{ bits/symbol}$$

c) explain in one sentence what entropy means

measure of randomness / uncertainty

Problem 2.2 :

a) compute the uncompressed data rate for stereo CD quality audio: 44.1 kHz sample rate, 16 bits per sample, 2 channels. express in mb/s

how many bits / second

$$\text{data rate}: \underbrace{44.1 \text{ kHz}}_{\text{sample/sec}} \cdot 16 \frac{\text{bits}}{\text{sample}} \cdot 2 \text{ channels} = 1{,}411{,}200 \frac{\text{bits}}{\text{second}} \longrightarrow 1.41 \frac{\text{Mb}}{\text{s}}$$

b) a streaming service sends 1080 videos at 8 Mb/s. the uncompressed rate for 1080p at 30 fps is 1.5 Gb/s. uncompressed ratio:

$$\frac{1.5 \frac{\text{Gb}}{\text{s}}}{8 \frac{\text{Mb}}{\text{s}}} = \frac{1500}{8} = 188 : 1$$

c) why is it impractical to stream uncompressed video over a typical home internet connection?

raw data requires high data rates, so compression is needed to make streaming possible

Problem 2.3

a) a hard drive is advertised as '2TB' using SI prefixes → how many bytes is this

$$2 \times 10^{12} \text{ B}$$

uses binary not decimal

b) an operating system reports the same drive as 1.82 TiB (tebibytes). are these consistent?

$$1.82 \cdot 2^{40} = 2 \cdot 10^{12} \text{ bytes} \longrightarrow \text{same}$$

c) a network link operates at 1 Gb/s. how long does it take to transmit 1 GiB of data

$$1 \frac{\text{Gb}}{\text{s}} \longrightarrow \frac{10^9 \text{ bits}}{\text{sec}} \quad : \quad 1 \text{ GiB} \longrightarrow 2^{30} \text{ Bytes} \longrightarrow 2^{30} \cdot 8 \text{ bits}$$

$$\frac{2^{30} \cdot 8 \text{ bits}}{10^9 \text{ bits/sec}} = 8.56 \text{ seconds}$$